Heat Exposure / Heat Exhaustion / Heat Stroke

Primary Survey

Consider non-environmental causes of hyperthermia

Cool Patient: "Strip, Spray, Fan, Ice"

- Strip: Remove clothing and loosen restrictive garments
- Spray: Tepid water over the patient.
- Fan: Move patient to a well ventilated, cool area.
- Ice: Apply cold packs to the axilla, groin, and posterior neck

Heat Exhaustion

- Elevated body temperature
- Headache
- Nausea /vomiting
- Syncope
- Heat cramps

Consider PO Fluids with Electrolytes

Consider IV Crystalloid
Adult: 500ml bolus
Pediatric: 10ml/kg bolus
Repeat as required

EMT

Heat Stroke

- CNS involvement (ataxia, coma, confusion, irritability, seizures), or
- Temperature usually >40°C, or
- Hot / Dry skin, or
- Hypotension

Consider ALS Back-Up

Consider IV Crystalloid
Adult: 500ml bolus
Pediatric: 10ml/kg bolus
Repeat as required

EMT

Evaporative Cooling

- Fan the patient
- Mist skin with tepid water

Consider Alternative Causes for altered mental status

Patient Care Goals

1. Rapidly identify and manage suspected heat exposure or hyperthermia to prevent progression to severe complications like heat stroke.
2. Provide immediate cooling and supportive care to reduce core body temperature and minimize secondary injury.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms suggestive of heat exposure or hyperthermia, including:
  - Exposure to hot environments (e.g., high ambient temperatures, exertion in heat).
  - Clinical features such as elevated body temperature, headache, nausea/vomiting, syncope, heat cramps, CNS involvement (e.g., ataxia, confusion, seizures), hot/dry skin, or hypotension.
  - Core temperature above 37°C, often >40°C in heat stroke, as measured or estimated.
  - History of prolonged heat exposure, physical exertion, or inadequate hydration in hot conditions.

Exclusion Criteria

- Hyperthermia or fever secondary to an intrinsic cause such as sepsis.

Key Patient Assessment Points

1. **Primary and Secondary Survey**
   - Perform primary and secondary surveys as per Universal Care CPG.
   - Consider non-environmental causes of hyperthermia (e.g., infections, medications).
2. **Differentiation of Conditions**
   - Heat Exhaustion: Elevated body temperature (typically 37–40°C), headache, nausea/vomiting, syncope, heat cramps, poor circulation, dehydration.
   - Heat Stroke: Core temperature usually >40°C, neurological dysfunction (delirium, confusion, seizures, loss of consciousness), hot/dry skin, hypotension; recognize exertional (from activity) vs. non-exertional (vulnerable populations like elderly).
3. **Severity Grading**
   - Mild (Heat Cramps/Exhaustion): Localized symptoms without CNS involvement.
   - Severe (Heat Stroke): Systemic effects with CNS dysfunction, potential for multi-organ failure.
4. **Monitoring**
   - Assess for dehydration, electrolyte imbalances, and altered mental status.
   - Monitor ECG for arrhythmias secondary to electrolyte imbalances.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary & Secondary Survey**
   - Perform Primary and Secondary Surveys as per Universal Care CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
3. **Initial Care**
   - Remove patient from hot environment and prevent further heat exposure.
4. **Cooling Interventions**
   - General Cooling:
     - Strip/remove clothing and loosen restrictive garments; spray/mist skin with tepid water; fan to promote evaporation; apply ice/cold packs to axilla, groin, and posterior neck.
   - Heat Exhaustion:
     - Passive cooling (remove from heat, fan, remove clothing).
   - Heat Stroke:
     - Aggressive active cooling as priority ("cool first, transport second")

        - Aim for cooling rate of 0.15°C/min or more.
        - Target reduction to 38.6–39.4°C to avoid overcooling.
        - Methods include:
            - Cold water immersion (gold standard if available), or/
            - Ice packs/fans during transport.

5. **PO Fluids and Electrolytes**
    - Indications:
        - Heat exhaustion (providing the patient is alert and able to swallow).
6. **IV Fluid**
    - Use Normal Saline or Lactated Ringers.
    - Indications:
        - Heat Exhaustion if signs of dehydration or shock.
        - Heat Stroke: Signs of heat stroke.
    - Dose:
        - Adult: 500 mL IV/IO bolus. (repeat as required, maximum 200mL).
        - pediatric: 10 mL/kg bolus (repeat as required, maximum 40mL/kg).
7. **Seizure Management**
    - Manage seizures or agitation with benzodiazepines as per Seizures CPG.
8. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- None Noted

Clinical Insights & Additional Guidance

- Progression Risk
    - Heat exhaustion can rapidly deteriorate to heat stroke if untreated; focus on rehydration and gentle cooling.
- Heat Stroke Emergency
    - This is a life-threatening condition with high risk of arrhythmias, electrolyte derangements, seizures, and multi-organ failure; aggressive treatment is required.
- Non-Environmental Hyperthermia
    - Consider intrinsic causes (e.g., malignant hyperthermia, drug toxicity) and manage accordingly.
- Special Populations
    - Elderly, children, and those with comorbidities are at higher risk; exertional heat stroke common in athletes.
- Antipyretics
    - Avoid antipyretics (e.g., acetaminophen) as they are ineffective; focus on external cooling.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Description of heat exposure (e.g., environment, exertion), duration.
- Injuries / Information: Initial temperature (if measured), clinical features, non-environmental causes considered.
- Signs: Vital signs, neurological status, skin condition, severity classification.
- Treatment:
    - Cooling methods (e.g., strip/spray/fan/ice, immersion).
    - Fluids administered (PO/IV, type, volume).
    - Any additional interventions (e.g., benzodiazepines for agitation).
    - Hospital notification.
- Allergies: Known allergies to medications.
- Medications: Document any administered drugs.
- Background: Relevant history (e.g., prior heat illness, comorbidities like heart disease).
- Other:

- Accurate timeline of exposure and symptom onset.
- Communication with receiving facility.

Key Performance Indicators

1. Assess and document body temperature.

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Pediatric Guidelines